



Alimentary system module

Liver, pancreas and biliary system

SGL 5

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Laboratory assessment of Liver functions

- laboratory assessment of liver integrity can be done through testing of three different elements of the liver structures and functions:

1. Hepatocyte integrity: is tests the cytosolic hepatocyte enzymes which includes:

1. Serum aspartate aminotransferase (AST).
2. Serum alanine aminotransferase (ALT)
3. Serum lactate dehydrogenase (LDH).

These enzymes will elevate in case when there is destruction of hepatocyte, e.g. by viruses.

Laboratory assessment of Liver functions

2. Biliary excretory function: it includes:

- Substances normally secreted in bile: Serum bilirubin
 - Total: unconjugated plus conjugated
 - Direct: conjugated only
- Urine bilirubin
- Serum bile acids
- Plasma membrane enzymes (from damage to bile canaliculus):
 - Serum alkaline phosphatase.
 - Serum g-glutamyl transpeptidase (GGT).

These tests will elevate in case when there is destruction/ obstruction to biliary passage.

Laboratory assessment of Liver functions

3. Hepatocyte synthetic function

- Proteins secreted into the blood.
 - Serum albumine
 - Coagulation factors
 - Prothrombin time (PT)
 - Partial thromboplastin time (PTT)
- Hepatocyte metabolism
- Serum ammonia
- Aminopyrine breath test (hepatic demethylation)

Liver Failure

- Is the inability of the liver to perform its normal synthetic, metabolic and excretory functions.
- Three types are present:
 1. Acute liver failure.
 2. Chronic liver failure and liver cirrhosis.
 3. Acute-on-chronic liver failure: in which a sudden acute liver failure develops in patient known to have years of stable, well-compensated, chronic liver disease. This usually follow a recent superimposed insult to the liver.

Acute Liver Failure

- Is defined as an acute liver illness that produces hepatic encephalopathy within 6 months of the initial diagnosis.
- Causes:
 - In the US, accidental or deliberate ingestion of acetaminophen accounts for almost 50% of cases of acute liver failure.
 - In Asia, acute hepatitis B and E predominate as causes of acute liver failure.
 - Acute hepatitis A.
 - Autoimmune hepatitis.
 - Other drugs and toxins

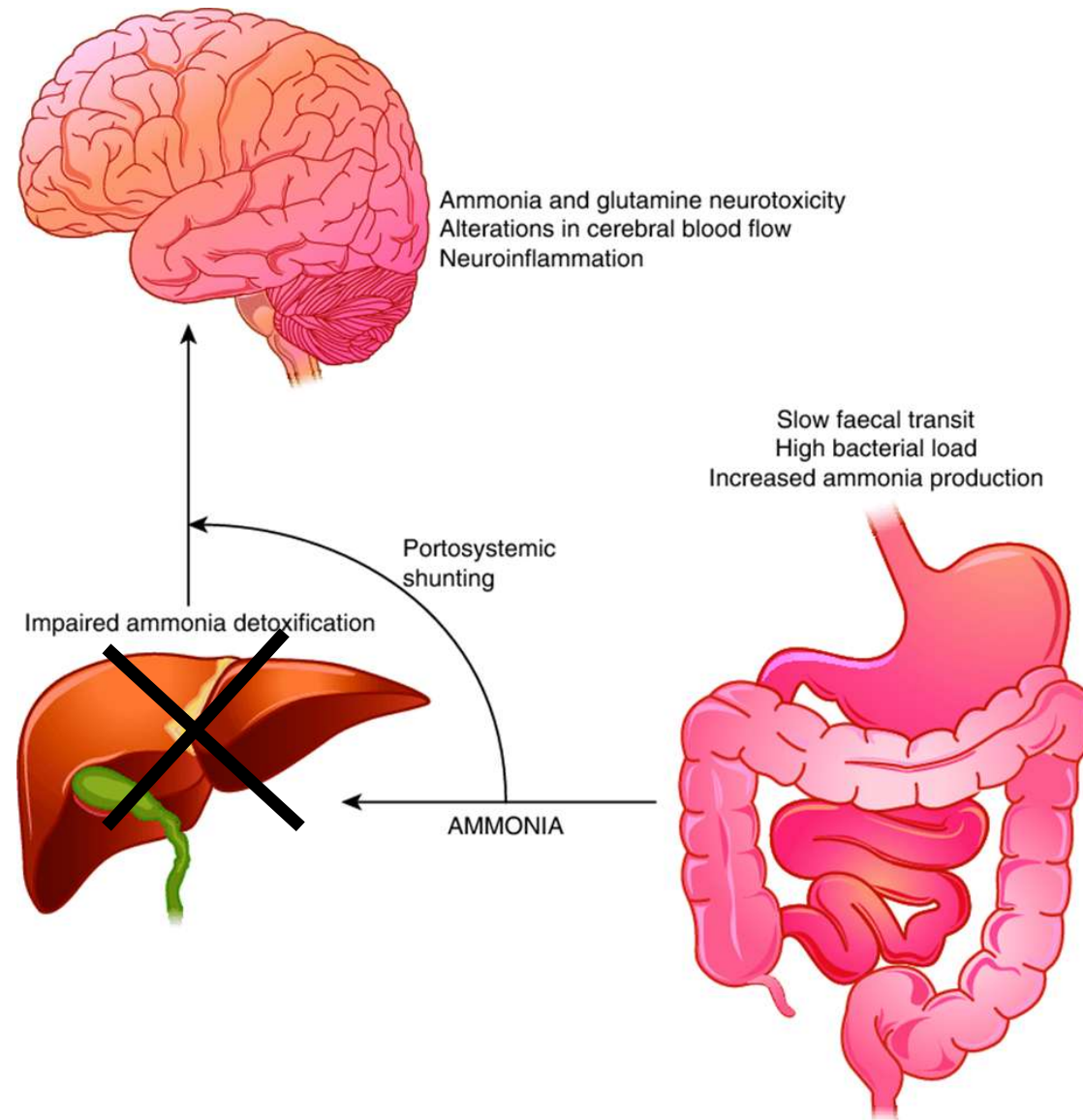
Acute Liver Failure

- **Pathology:** characterized by massive hepatic necrosis.
 - **Gross pathology,** the liver is small and shrunken due to loss of parenchyma.
 - **Microscopically,** there are large zones of destruction surrounding occasional islands of regenerating hepatocytes while scarring is mostly absent because of the acute nature of the process.
- **Clinical Features:** nausea, vomiting, jaundice, and fatigue, which are followed by the onset of life-threatening encephalopathy, coagulation defects, and portal hypertension associated with ascites.

Acute Liver Failure

Hepatic encephalopathy:

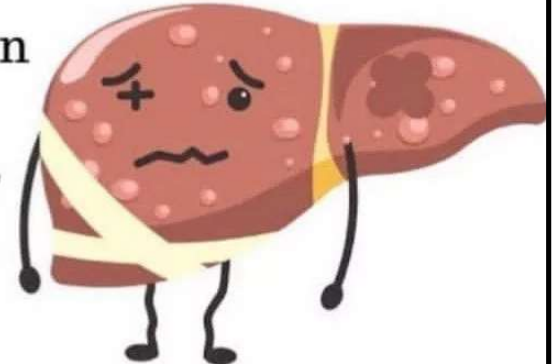
- It is characterized by various symptoms ranging from subtle behavioral abnormalities to confusion, stupor, coma, and death. A typical neurologic sign is asterixis
- It is believed to be caused by elevated ammonia levels, which correlate with impaired neuronal function and cerebral edema.



Several condition participate hepatic encephalopathy and these includes:

mnemonic: **HEPATICS**

H hemorrhage in GIT
E excess protein in diet
P potassium depletion (diuretics)
A alkalosis, azotemia
T trauma
I infections (bacterial, peritonitis etc)
C constipation (colon surgery)
S sedatives

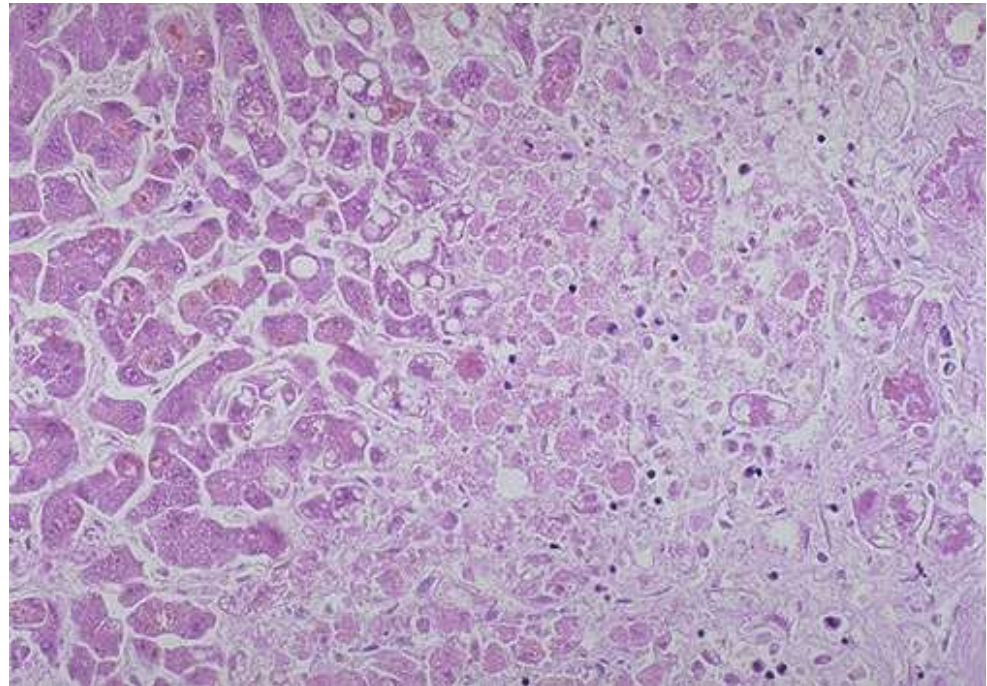


Acute Liver Failure

Massive liver necrosis, the liver is small, bile stained, soft, and congested.



Massive liver necrosis in acetaminophen overdose: There is extensive hepatocyte necrosis seen here in while those at the left are dying.



Chronic Liver Failure and Cirrhosis

- Cirrhosis refers to the diffuse transformation of the liver into regenerative parenchymal nodules surrounded by fibrous bands.
- It is not a specific entity, but a morphologic change often associated with severe chronic liver disease.
- Causes:
 - Chronic hepatitis B, chronic hepatitis C
 - Nonalcoholic fatty liver disease (NAFLD)
 - Alcohol-related liver disease.
- Cirrhosis It has variable prognostic outcomes.
- Not all chronic liver disease ends in cirrhosis.
- Not all cirrhosis leads to end-stage liver disease.

Liver Cirrhosis/ Clinical Features and consequences:

- About 40% of individuals are asymptomatic.
- Nonspecific manifestations, as anorexia, weight loss, weakness.
- Signs and symptoms of liver failure as Jaundice, encephalopathy, and coagulopathy etc.
- Chronic severe jaundice can lead to pruritus (itching).
- Portal hypertension from increased vascular resistance coupled with increased portal blood flow.
- Portosystemic shunts develop due to sustained portal hypertension which leads to dilation of collateral vessels wherever the systemic and the portal circulations share common capillary beds; e.g. Esophagogastric varices, caput medusa, rectal varices/ hemorrhoides.
- Ascites of transudate type.
- Splenomegaly and hypersplenism.
- Hyperestrogenemia which lead to development of angiomas, gynaecomastia and others.
- Predispose to development of hepatocellular carcinoma

Chronic Liver Failure and Cirrhosis/ Pathologic Features:

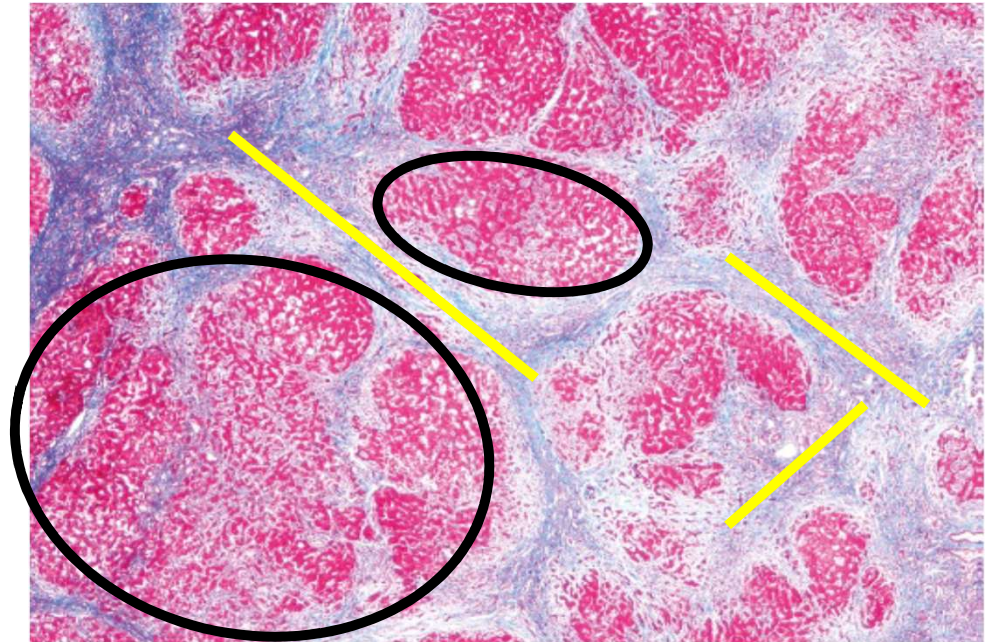
Gross pathology:

Multiple regenerative nodules throughout the entire liver, separated by bridging fibrosis.



Microscopic pathology:

Diffuse disruption in architecture of the entire liver (loss of normal central - portal relationship)
Bridging fibrous septa (Yellow lines).
parenchymal nodules of regenerating hepatocytes (black circles).



Liver tumors

Liver tumors

- The liver is a site of different benign and malignant, primary and metastatic, tumors.
- The primary liver tumors arise from different cellular elements of the liver like hepatocyte, biliary tree, vascular element, ...etc.
- The clinical presentation may range from asymptomatic, detected on abdominal imaging, to those of abdominal pain, abdominal mass or impaired liver functions e.g. obstructive jaundice.

Benign liver tumors

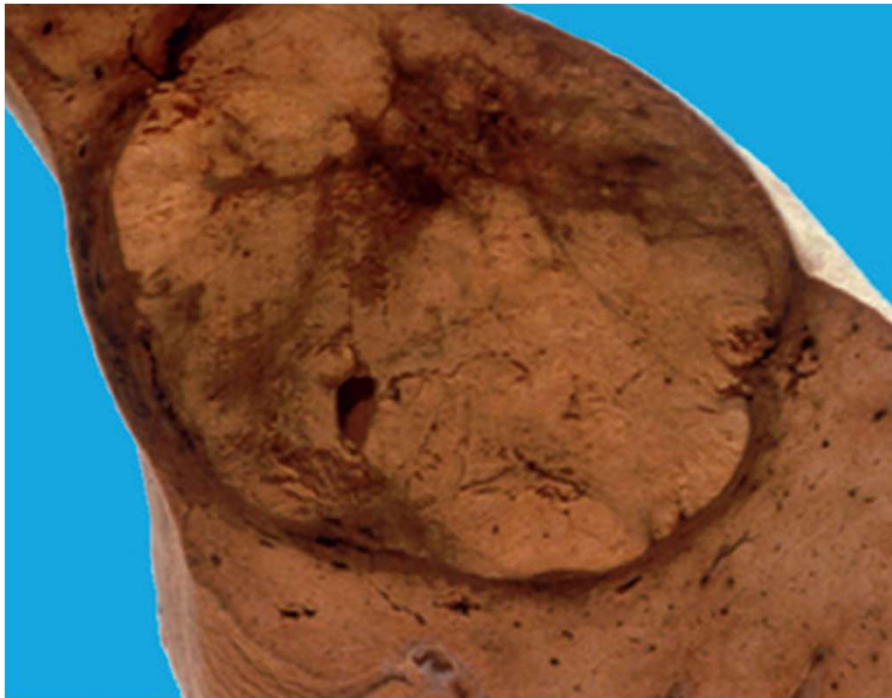
Hepatocellular adenoma:

- Is a benign tumor that usually arises in a non-cirrhotic liver in reproductive-age women.
- Aetiology:
 - Associated with the use of oral contraceptive drugs.
 - Obesity and metabolic syndrome.
 - Certain genetic mutation, e.g. B catenin.
- Clinical significance:
 - Prone to rupture and causes intra-abdominal hemorrhage, this possibility increase with increase in tumor size.
 - Risk of malignant transformation.

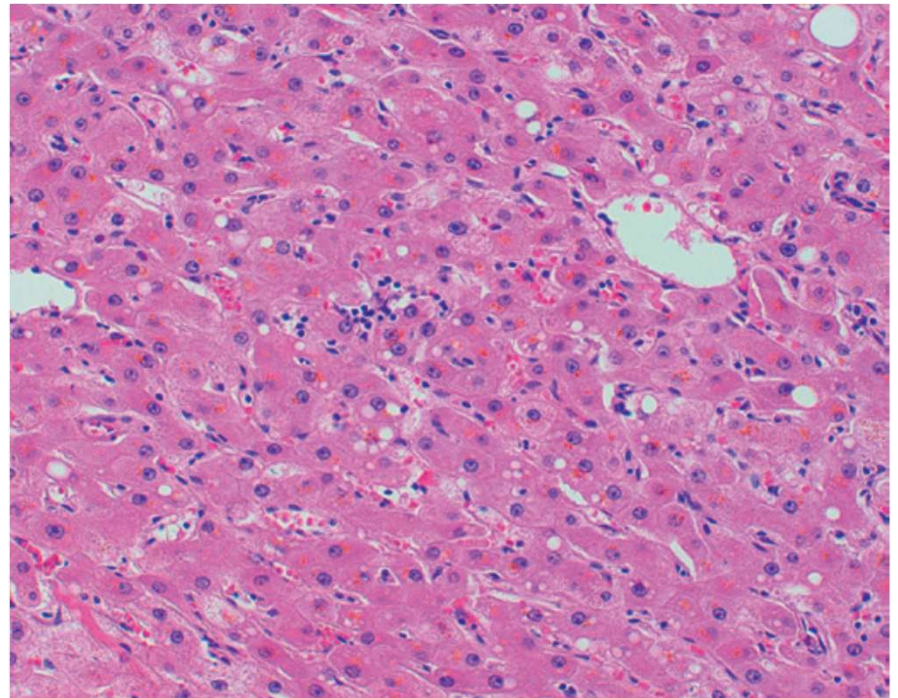
Benign liver tumors

Hepatocellular adenoma:

A well demarcated liver mass, similar to background liver tissue.



The tumor cells are uniform with minimal cytological atypia and are arranged in architecturally normal hepatic plates



Benign liver tumors

Hepatic hemangiomas:

- Are benign vascular tumors.
- Are the most common benign tumor of the liver.
- Could be solitary or multiple.
- Clinical significance:
 - The majority are asymptomatic with no consequences.
 - Larger lesions, more than 5cm are liable for:
 - Hemorrhage.
 - Thrombosis and consumptive coagulopathy.

Malignant liver tumors

Hepatocellular Carcinoma (HCC):

- Is the commonest primary malignant tumor of the liver in adults.
- Clinically: patient presents with features of underlying chronic liver diseases, if present, abdominal pain, malaise, fatigue, weight loss, and sometimes awareness of an abdominal mass.
- Serum levels a-fetoprotein is used as tumor marker, however, it is elevated in 50% of individuals with advanced HCC, but this is neither a sensitive nor a specific marker.

Malignant liver tumors/ HCC

Aetiology:

- 1. Chronic HBV infection: HCC is highest in Asia (southeast China, Korea, Taiwan) and sub-Saharan Africa, in which HBV is transmitted vertically.**
- 2. Hepatitis C infection: is an increasingly cause for HCC in western world.**
- 3. Aflatoxin: a mycotoxin produced by Aspergillus species that contaminates staple food crops in Africa and Asia.**
- 4. Alcohol.**
- 5. NAFLD.**
- 6. Inherited disorders, particularly hereditary hemochromatosis and α 1AT deficiency, and, to a lesser degree, Wilson disease.**

Malignant liver tumors/ HCC

Pathology:

- **Grossly: HCC may grow as:**
 - **A unifocal (usually large) mass.**
 - **Multifocal, widely distributed nodules of variable size.**
 - **A diffusely infiltrative cancer.**
- **Microscopically: HCC may have different morphology ranging from well differentiated, simulating normal liver tissue, to high grade pleomorphic tumor.**

- Hepatocellular carcinoma. A fairly circumscribed cream-tan, solid mass with hemorrhage in a cirrhotic liver background



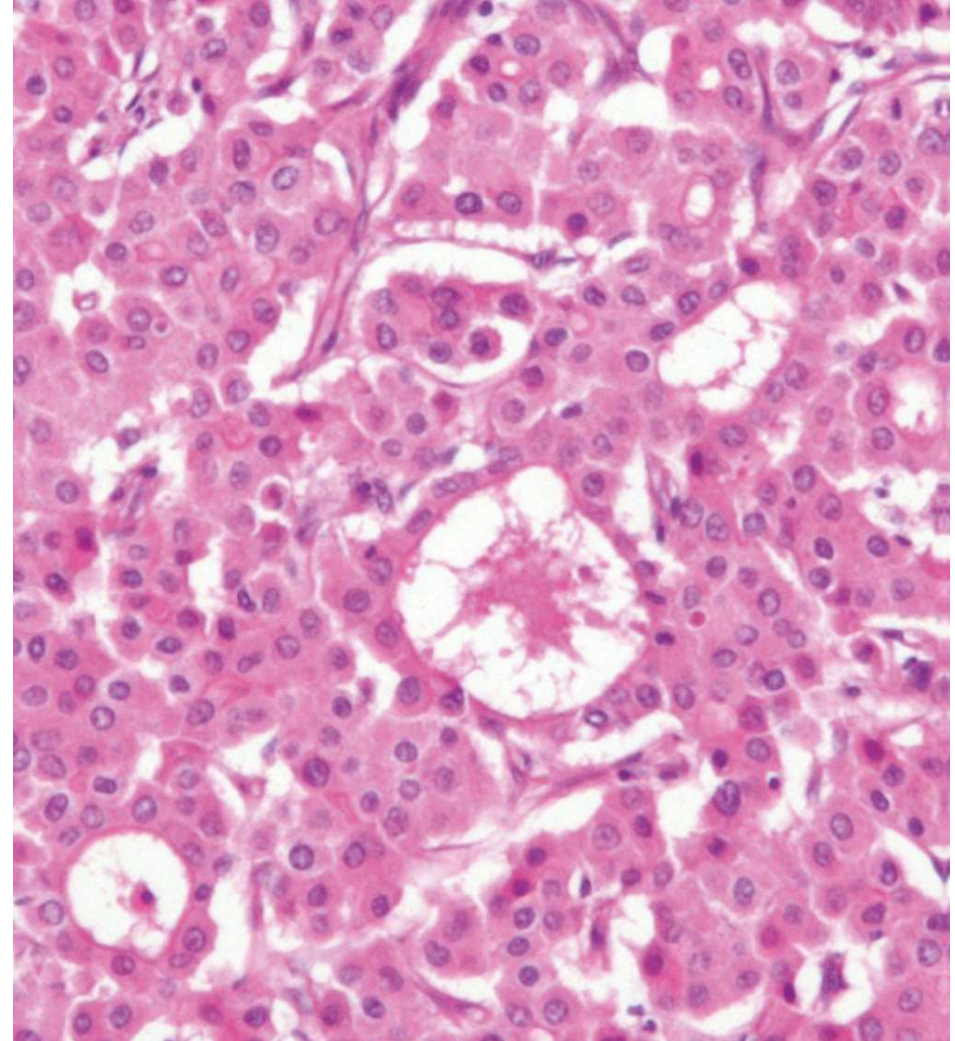
- Hepatocellular carcinoma. Solid, multiple, expanding tumor nodules with variegated light yellow to brown appearance and punctuated with hemorrhage. The background liver is cirrhotic



Hepatocellular carcinoma (diffuse pattern).
The tumor is composed of hundreds of small light brown nodules that are grossly hard to differentiate from cirrhotic nodules.

HCC:

Malignant hepatocytes growing in distorted architecture: large pseudoacinar spaces, malformed, dilated bile canaliculi, and thickened hepatocyte trabeculae.



Malignant liver tumors/ Others

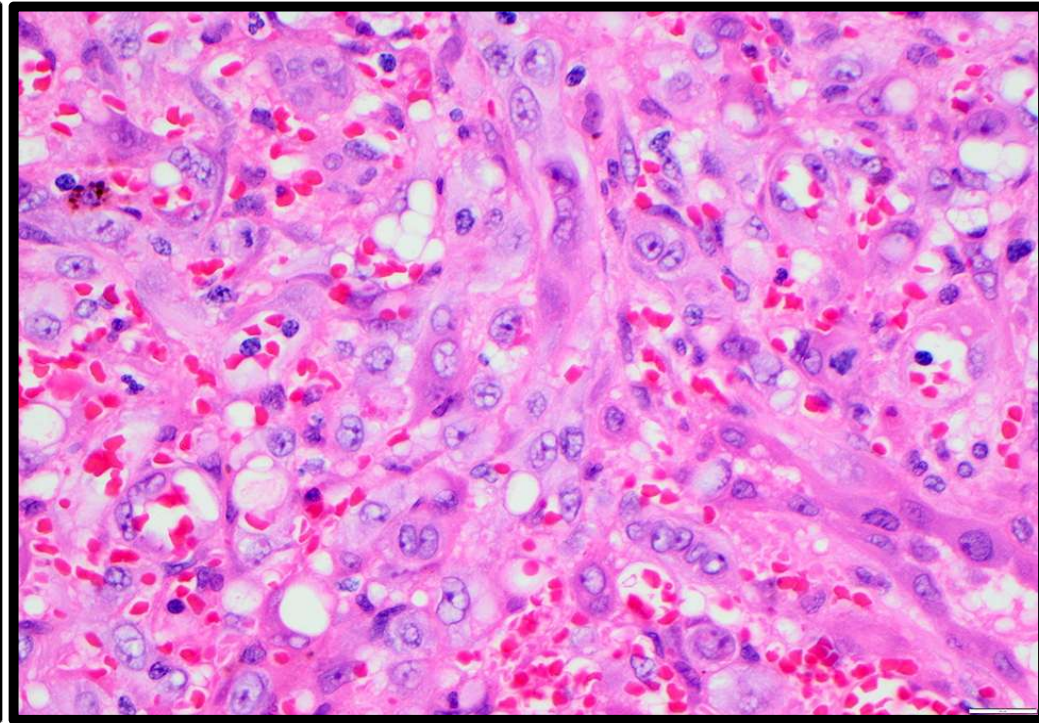
- **Hepatoblastoma**: is the most common hepatic malignancy in infants and children. Histology shows immature precursor cells of variable combination of epithelial and mesenchymal components.
- **Angiosarcoma**: malignant tumor of vascular origin, is the commonest primary sarcoma of the liver, associated with exposure to androgen steroids, arsenic, Thorotrast (previously used radiocontrast agent) and vinyl chloride.
- **Metastatic tumors**: are the most common malignancy found within the liver. Common primary sites include the colon, breast, and lung. Metastatic tumors tend to occur as multiple well-circumscribed masses.

Malignant liver tumors/ angiosarcoma

**Tumor form diffusely infiltrative,
hemorrhagic and blood filled cavities mass**



**Tumor composed of malignant cells form
anastomosing vascular channels**



Malignant liver tumors/ metastasis

- Metastatic tumors tend to occur as multiple well-circumscribed masses.



References

- **Robbins & Cotran Pathologic Basis of Disease - 10th edition -2021.**
- **Robbins Basic Pathology - 11th edition 2022.**
- **web**